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KasSigner

Quick Start Guide

Air-gapped offline signing device for Kaspa.
From unboxing to your first signed transaction.

Release v1.0.7

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Licence GPL-3.0

Repository github.com/InKasWeRust/KasSigner

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KasSigner is an EXPERIMENTAL offline signing device. It is NOT a hardware wallet. It has no secure element and no persistent storage; all keys are wiped on power-off. It has not been professionally audited. Do not use it to manage funds you cannot afford to lose.

Don't trust, verify. Sign offline. Broadcast sovereign.

1. What You Need

	Waveshare ESP32-S3-Touch-LCD-2	M5Stack CoreS3 Lite
Price	~$25	~$45
Display	ST7789T3 320x240	ILI9342C 320x240
Camera	OV2640 / OV5640 DVP (auto-detect)	GC0308 QVGA
SD Card	SDHOST native (fast)	Bitbang SPI
Best for	QR scanning quality	Portability, speaker

You also need a computer (macOS, Linux, or Windows), a USB-C data cable (not charge-only), a microSDHC card (4 to 32 GB, FAT32) for backups, and a terminal.

2. Install the Build Toolchain

```
# Step 1: Install Rust
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source ~/.cargo/env

# Step 2: Install espup + espflash
cargo install espup && espup install
cargo install espflash

# Step 3: Clone and verify
git clone https://github.com/InKasWeRust/KasSigner.git
cd KasSigner/tools && cargo run --bin kassigner-setup && cd ..
```

If you would rather verify than trust, build in Docker instead: docs/REPRODUCIBLE_BUILD.md. Same firmware, and a hash you can compare against the release.

3. Build and Flash

```
# Waveshare (OV5640 with AF module; drop ov5640-af for a plain module, use ov2640-wide for OV2640):
cd bootloader
ESP_HAL_CONFIG_PSRAM_MODE=octal cargo run --release --features ov5640-af

# M5Stack CoreS3 Lite:
cargo run --release --no-default-features --features m5stack
```

The device flashes, reboots, runs its self-tests and cryptographic known-answer tests (about two seconds, **All crypto KATs passed** on the serial log) and shows the home screen.

4. Create Your First Wallet

- Tap **Tools**, **Seed Tools**, then **New Seed** (camera, plus IMU on Waveshare), **Dice Seed** (99 or 198 rolls of a real die, fully verifiable) or **Touch Seed** (your own scribble); choose 12 or 24 words
- Camera mode refuses a dark or still scene with **Need more light**: point at something lit and moving
- Words are shown one at a time; tap to advance

> **WARNING:** These words ARE your wallet. Write them down physically. Do not photograph them. Do not type them into any internet-connected device.

Optional BIP39 passphrase (25th word): words alone lead to a decoy wallet, words + passphrase to the real one. The device never stores the passphrase; keep it apart from the words.

5. Sign Your First Transaction

- Export kpub: **Seeds**, **Export**, **Watch-Only**, **kpub as QR**
- Import it into KasSee Web (kassigner.org) by scanning the QR or pasting it
- In KasSee, enter destination and amount, tap Send: an animated QR with the unsigned transaction
- On the device: **Scan**, review every output, amount and fee, tap **CONFIRM**
- The device shows the signed transaction as a QR
- In KasSee, **Broadcast**, scan the signed QR: the transaction is submitted

6. Back Up Your Seed

- **Encrypted SD**: Seeds, Export, Seed Backup, Backup to SD, choose a password. Restore: Tools, Import / Export, Import from SD
- **Steganographic JPEG** (recommended): Seeds, Export, Steganography, pick a photo, type a descriptor (it is the password). Run it once as Descriptor and once as Picture with the same descriptor to cover both metadata stripping and re-compression
- **SeedQR (paper)**: Seeds, Export, Seed Backup, QR Export. Unencrypted; always use a passphrase

Whatever you choose, import it back on the device once before you trust it.

7. Tips and Features

- Multiple seeds: up to 16 slots in RAM. Tap to switch. All wiped after four minutes idle, and on power-off
- BIP85 child seeds: Tools, Seed Tools, BIP85 Child
- Multisig (45'): each cosigner exports **kpub Multisig QR**; one device creates the wallet under Tools, Multisig; back up seed AND descriptor
- Message signing: Tools, Single Signature, Sign Message
- kpub to SD: save or import kpub files from the card
- Reproducible builds: Docker, bit-identical on any machine
- Secure Boot: burn the eFuses per docs/EFUSE_RUNBOOK.md when you are ready to commit a board

8. Troubleshooting

- Stuck or frozen: unplug USB, wait 10 seconds, reconnect
- QR not scanning: bright even light, 10 to 15 cm, hold steady, tilt 20 to 45 degrees against reflections; dense codes closer, small codes a little further
- SD not detected: FAT32 only, microSDHC recommended. Settings, SD Card, Format
- Signing failed: check the right seed is loaded and active (fingerprint); the signer takes up to 32 inputs and 8 outputs, consolidate in KasSee if needed
- Touch unresponsive: taps fire on release; a firm, deliberate tap

9. Command Reference

```
# Clone
git clone https://github.com/InKasWeRust/KasSigner.git && cd KasSigner

# Environment check
cd tools && cargo run --bin kassigner-setup && cd ..

# Build + flash (Waveshare, OV5640 AF)
cd bootloader
ESP_HAL_CONFIG_PSRAM_MODE=octal cargo run --release --features ov5640-af

# Build + flash (Waveshare, OV2640 wide-angle)
ESP_HAL_CONFIG_PSRAM_MODE=octal cargo run --release --features ov2640-wide

# Build + flash (M5Stack CoreS3 Lite)
cargo run --release --no-default-features --features m5stack

# Docker reproducible build
docker build --platform linux/amd64 -f Dockerfile.base -t kassigner-toolchain:v3 .
docker build --platform linux/amd64 -t kassigner-build . && docker run --rm kassigner-build

# KasSee Web (WASM build)
cd kassee && RUSTUP_TOOLCHAIN=stable ./build.sh
```

> What happens when you power off? Everything is gone: every key, every seed, every passphrase, wiped from RAM. The device boots fresh every time. Your wallet lives in your backup; the device is the tool that brings it to life, signs what needs signing, and forgets.